: smithlindsay.github.io

Lindsay M. Smith

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Research Interests

My research focuses on the **science of AI** and **AI for science**. I apply techniques and ideas from physics to artificial neural networks to understand how they learn, and I develop methods for AI to learn to conduct science effectively. My current research in machine learning and AI includes projects in **interpretability, AI for scientific discovery, in-context learning, LLM multi-agent interactions, and AI safety**.

Education

M.A., Ph.D. — Princeton University — Physics 2022 - 2027

Advisors: William Bialek and David Schwab.

NSF GRFP

B.A. — University of Pennsylvania — Physics (Honors) 2018 - 2022

Advisor: Dani Bassett.

Cum Laude; Minors in Mathematics and French & Francophone Studies.

Experience

Junior Research Scientist — Polymathic AI Mar 2026 - Aug 2026

Internship Project: AI for Scientific Discovery (NeurIPS 2026 paper and paper under review at ICLR 2027).

Mentee — MARS (Mentorship for Alignment Research Students) Jul 2025 - Feb 2026

MARS 3.0 participant at the Cambridge AI Safety Hub. Co-first author on project investigating AI control and steering via Chain-of-Thought (CoT) injections (paper at workshop at ICLR 2026).

Graduate Researcher — Princeton University Aug 2022 - Present

Awards & Honors

Citadel GQS Physics PhD Fellowship Finalist (3 finalists nationally) 2026

National Science Foundation (NSF) ACCESS Computing Award 2025

(200,000 credits, $\sim$ $5,000)

Applied for and am Principal Investigator for this award.

NSF AI Institutes Virtual Organization (AIVO) Travel Grant ($2,000) 2025

American Physical Society (APS) GSNP Student Speaker Award Finalist ($500) 2025

NSF Graduate Research Fellowship Program (GRFP) ($159,000 + $12,000 bonus from Princeton) 2022 - 2027

Charlotte and Morris Tanenbaum *52 Graduate Fellowship in Physical or Life Sciences ($40,000) 2022 - 2023

Joseph Henry Merit Award ($5,000) 2022

University Scholars Program ($7,000) 2020 - 2022

Applied for and awarded summer research funding in 2020 and 2021.

National French Honor Society – Pi Delta Phi 2020 - Present

Sister Loretta Thome Scholarship ($10,000) 2018 - 2023

Publications [Google Scholar] **Equal contribution indicated by** $*$

10. **Lindsay M. Smith***, Matt L. Wiemann*, Peter Melchior, Salman Rahman, Andrew Gordon Wilson, Pavel Izmailov, Carolina Cuesta-Lázaro (2026). Discover-Many-Worlds: Reinforcement Learning for Scientific Discovery in Automatically Generated Physics. *Under review at ICLR 2027.*

9. Matt L. Wiemann*, **Lindsay M. Smith***, Peter Melchior, Siddharth Mishra-Sharma, Andrew Gordon Wilson, Pavel Izmailov, Carolina Cuesta-Lázaro (2026). DiscoverPhysics: Benchmarking LLMs for Out-of-the-Box Scientific Thinking. *Accepted at **NeurIPS 2026** Evaluations and Benchmarks*, https://arxiv.org/abs/2605.26087.

8. **Lindsay M. Smith***, Ananya Malik*, Edward James Young, Puria Radmard, Cameron Tice, Hannes Whittingham (2026). Chain-of-Thought Injection as an Inference-Time Safety Intervention. ***ICLR 2026** Workshop on Logical Reasoning of LLMs*, https://openreview.net/forum?id=v0XkjgeD6U.

7. Alireza Karami, Veronica Chelu, Po-Chen Kuo, **Lindsay M. Smith**, Mia Whitefield, Jonathan D. Cohen (2026). A Computational Basis of Natural Intelligence. *Proceedings of the Analytical Connectionism Schools 2023–2024, **PMLR** 320:151-176*, https://proceedings.mlr.press/v320/karami26a.html.

6. Colin Scheibner*, **Lindsay M. Smith***, William Bialek (2025). Large language models and the entropy of English. *Under review at Physical Review Letters*, https://arxiv.org/abs/2512.24969.

5. Jeff Shen* & **Lindsay M. Smith*** (2025). Alice: An Interpretable Neural Architecture for Generalization in Substitution Ciphers. *Preprint*, https://arxiv.org/abs/2509.07282.

4. Chase Goddard, **Lindsay M. Smith**, Vudtiwat Ngampruetikorn*, David J. Schwab* (2025). When can in-context learning generalize out of task distribution? ***ICML 2025***, https://arxiv.org/abs/2506.05574.

3. **Lindsay M. Smith**, Chase Goddard, Vudtiwat Ngampruetikorn*, David J. Schwab* (2024). Model Recycling: Model component reuse to promote in-context learning. ***NeurIPS 2024** Workshop on Scientific Methods for Understanding Deep Learning,* https://openreview.net/forum?id=vWSu8nEURM.

2. Chase Goddard, **Lindsay M. Smith**, Vudtiwat Ngampruetikorn*, David J. Schwab* (2024). Specialization-generalization transition in exemplar-based in-context learning. ***NeurIPS 2024** Workshop on Scientific Methods for Understanding Deep Learning,* https://openreview.net/forum?id=D1ui5QwHqF.

1. **Lindsay M. Smith**, Jason Z. Kim, Zhixin Lu, Dani S. Bassett (2022). Learning continuous chaotic attractors with a reservoir computer, ***Chaos*** 32, 011101, https://doi.org/10.1063/5.0075572. *Selected as an Editor's Pick and publicized with a Scilight summary: https://doi.org/10.1063/10.0009079.*

Talks

Kempner Institute Workshop: Learning Dynamics in Natural & Artificial Intelligence, Cambridge, MA — Sep 2026
Talk: "Learning long-ranged structure in English: Code length and entropy dynamics during LLM training."

APS Global Summit, Denver, CO — Mar 2026
Talk: "Universality of LLM mechanisms across scale and diversity."

Princeton University Machine Learning Journal Club, Princeton, NJ — Sep 2025
Invited talk: "Alice: An Interpretable Neural Architecture for Generalization in Substitution Ciphers."

APS Global Summit, Anaheim, CA — Mar 2025
(GSNP Student Speaker Award Finalist)
Talk: "Multi-Agent Debate: Analyzing Consensus in Networks of LLM Agents."

APS March Meeting, Chicago, IL — Mar 2022
Talk: "Learning Continuous Chaotic Attractors with a Reservoir Computer."

Posters

Princeton Center for Theoretical Science Workshop: The Physics of John Hopfield, Princeton, NJ — Sep 2025
"ALICE: An Interpretable Neural Architecture for Generalization in Substitution Ciphers."

Meta 2025 PhD Forum, Menlo Park, CA — Oct 2025
"ALICE: An Interpretable Neural Architecture for Generalization in Substitution Ciphers."

NeurIPS 2024 Workshop on Scientific Methods for Understanding Deep Learning, Vancouver, Canada — Dec 2024
"Model Recycling: Model component reuse to promote in-context learning."

Conference for Undergraduate Women in Physics (CUWiP), Virtual — Jan 2022
"Learning Continuous Chaotic Attractors with a Reservoir Computer."

Penn Research Expo, Philadelphia, PA — 2020, 2021, 2022
"Development of control in brain networks over temporal and spatial scales using graph models."
"Learning Continuous Chaotic Attractors with a Reservoir Computer."

Teaching and Mentoring

Teaching Assistant, Princeton University — Fall 2025
PHY 103: General Physics I

Mentor, Princeton Towards an Inclusive Physics cOmmunity in graduaTe Students (TIPOTS) — 2024 - Present
Mentored younger physics graduate students each semester.

Mentor, Princeton Physics Mentorship — 2022 - Present
Mentored one to four undergraduate physics students each semester. Met at least once a semester to give career and academic advice.

Tutor, CIS 110: Introduction to Computer Programming — 2021 - 2022
Tutored two to three students weekly.

Mentor, Side By Side Agency — 2021
Advised a student on an independent astrophysics research project and poster.

Professional Service and Outreach

Reviewer: ICLR 2027; Workshop on Scientific Methods for Understanding Deep Learning (Sci4DL) at NeurIPS 2024 and ICLR 2026.
Princeton Women in Physics, **Executive Board** (2023 - Present)
Princeton Physics EDI, **Events Committee** (2022 - 2024)

Technical Skills

Python, PyTorch, JAX, LaTeX, bash, Git, Jupyter, MATLAB, Java, C++, ROOT, Mathematica, Linux